Global Job Market Analysis Using Data Mining

Applying data mining techniques to uncover salary patterns, employment trends, and workforce insights from a global job market dataset.

Subject

DMBI

Team

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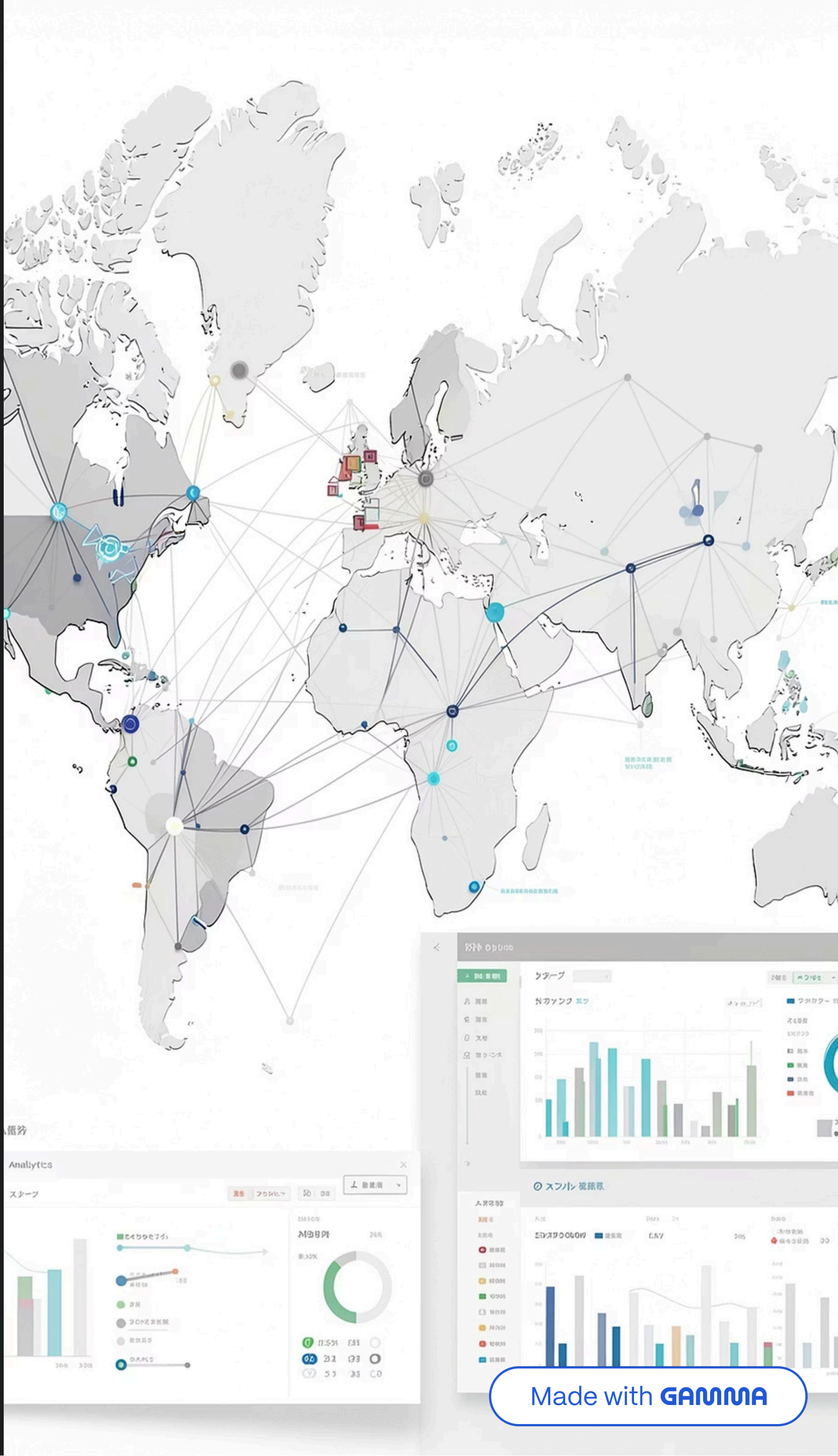
Roll No.

05, 06, 08, 09

Dataset

Job Market Dataset (Global)



Problem Statement

The modern job market is complex and data-rich. This project uses data mining to extract actionable insights from global employment data.

1 Analyze Global Trends

Examine how employment patterns vary across countries, industries, and time periods.

2 Understand Salary Factors

Identify which variables — experience, education, location — most influence compensation.

3 Predict Salaries

Apply machine learning models to forecast salaries based on key job attributes.

4 Discover Employment Patterns

Cluster job roles to reveal hidden groupings and structural trends in the workforce.

Dataset Description

500,000 SYNTHETIC JOB RECORDS

A1

	A	B	C	D	E	F	G	H	I
1		Job Id	Experience	Qualificati	Salary Ran	location	Country	latitude	longitude
2	0	1.09E+15	5 to 15 Yea	M.Tech	$59K-$99K	Douglas	Isle of Man	54.2361	-4.5481
3	1	3.98E+14	2 to 12 Yea	BCA	$56K-$116	Ashgabat	Turkmenis	38.9697	59.5563
4	2	4.82E+14	0 to 12 Yea	PhD	$61K-$104	Macao	Macao SAR	22.1987	113.5439
5	3	6.88E+14	4 to 11 Yea	PhD	$65K-$91K	Porto-Novo	Benin	9.3077	2.3158
6	4	1.17E+14	1 to 12 Yea	MBA	$64K-$87K	Santiago	Chile	-35.6751	-71.5429
7	5	1.17E+14	4 to 12 Yea	MCA	$59K-$93K	Brussels	Belgium	50.5039	4.4699
8	6	1.29E+15	3 to 15 Yea	PhD	$63K-$103	George To	Cayman Is	19.3133	-81.2546
9	7	1.5E+15	2 to 8 Year	M.Com	$65K-$102	SÃƒÂ£o Tor	Sao Tome	0.1864	6.6131
10	8	1.68E+15	2 to 9 Year	BBA	$65K-$102	Male	Maldives	3.2028	73.2207
11	9	2.56E+14	1 to 10 Yea	BBA	$60K-$80K	Saint John'	Antigua an	17.0608	-61.7964
12	10	2.7E+15	3 to 10 Yea	BCA	$57K-$104	Manama	Bahrain	26.0667	50.5577
13	11	1.45E+15	4 to 12 Yea	B.Tech	$64K-$98K	The City of	Bermuda	32.3078	-64.7505
14	12	1.91E+15	3 to 15 Yea	MCA	$65K-$122	Kingston	Jamaica	18.1096	-77.2975
15	13	2.91E+14	1 to 8 Year	B.Com	$56K-$86K	Banjul	Gambia	13.4432	-15.3101
16	14	1.63E+15	1 to 9 Year	MCA	$57K-$98K	Damascus	Syrian Arab	34.8021	38.9968
17	15	2.69E+15	4 to 12 Yea	M.Com	$65K-$100	Sanaa	Yemen	15.5527	48.5164
18	16	2.82E+15	5 to 14 Yea	M.Tech	$60K-$83K	San Marino	San Marino	43.9424	12.4578
19	17	7.69E+13	0 to 11 Yea	BA	$55K-$117	Papeete	French Pol	-17.6797	-149.407
20	18	2.35E+14	3 to 12 Yea	BA	$55K-$121	Seoul	North Kore	40.3399	127.5101

	J	K	L	M	N	O	P	Q	R
1	Work Type	Company	Job Posting	Preference	Contact Pe	Contact	Job Title	Role	Job Porta
2	Intern	26801	########	Female	Brandon C	001-381-9	Digital Mar	Social Mec	Snagajob
3	Intern	100340	########	Female	Francisco	461-509-4	Web Devel	Frontend V	Idealist
4	Temporary	84525	########	Male	Gary Gibso	9.69E+09	Operations	Quality Co	Jobs2Ca
5	Full-Time	129896	########	Female	Joy Lucero	+1-820-64	Network En	Wireless N	FlexJobs
6	Intern	53944	########	Female	Julie Johns	343.975.47	Event Man	Conferenc	Jobs2Ca
7	Full-Time	23196	########	Male	Matthew G	(973)791-5	Software T	Quality Ass	Snagajob
8	Temporary	26119	########	Both	Zachary Ha	001-268-5	Teacher	Classroom	FlexJobs
9	Contract	40558	########	Female	Chad Stric	667.202.68	UX/UI Desi	User Interf	Indeed
10	Temporary	105343	########	Female	David Han	+1-337-94	UX/UI Desi	Interaction	Indeed
11	Full-Time	102069	########	Both	John Nguye	001-318-9	Wedding P	Wedding C	Stack Ov
12	Contract	130338	########	Female	John Collie	001-683-87	QA Analyst	Performan	Glassdo
13	Contract	117285	########	Male	Tammy Ca	001-388-4	Litigation A	Family Law	Stack Ov
14	Part-Time	79071	########	Both	Jennifer Mc	200-851-9	Mechanica	Mechanica	Stack Ov
15	Temporary	127900	########	Female	Lisa Frankl	(229)625-8	Network A	Network Se	FlexJobs
16	Full-Time	92128	########	Male	Adam Whit	001-805-8	Account M	Sales Acco	USAJOBS
17	Part-Time	92028	########	Male	Michael Gr	001-609-5	Brand Man	Product Br	SimplyH
18	Part-Time	49100	########	Male	Samuel Pe	(434)560-6	Social Wor	School Soc	Glassdo
19	Full-Time	29318	########	Male	Benjamin L	(331)668-1	Social Mec	Content Cr	Jobs2Ca
20	Contract	126630	########	Female	Adam And	459-591-7	Email Mark	Deliverabil	Internsh

job_descriptions

Copilot

A large-scale synthetic dataset with no real personal data, designed to simulate realistic global job market conditions across multiple dimensions.

Location

Country & City

Job Role

Occupation & Field

Work Profile

Experience & Employment Type

Demographics

Education, Gender & Company Size

Compensation

Salary (target variable)

Time

Year & Month of record

Data Exploration (EDA — Part 1)

EXPLORATORY DATA ANALYSIS

500K

Total Records

Rows in the dataset

12

Features

Columns per record

$207K

Avg. Salary

Mean annual compensation

25

Max Experience

Years range: 0 – 25

No Missing Values

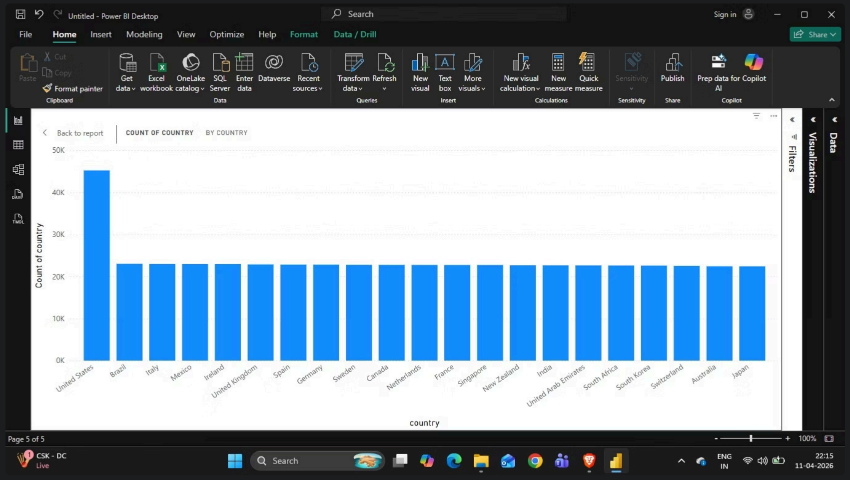
Dataset is complete — no imputation required, ensuring reliable downstream analysis.

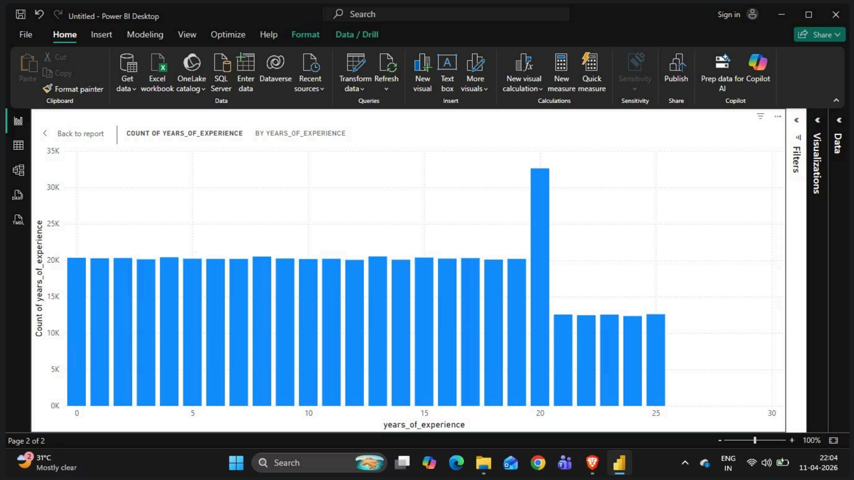
Mixed Data Types

Both categorical (occupation, country) and numerical (salary, experience) columns are present.

Well-Distributed

Records span multiple years, months, and geographies, supporting trend analysis.

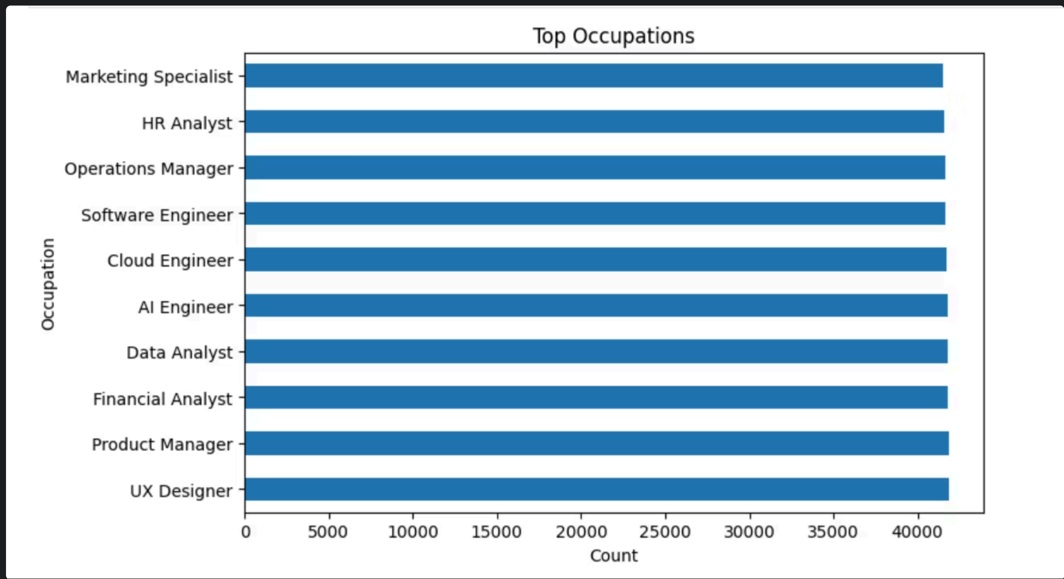


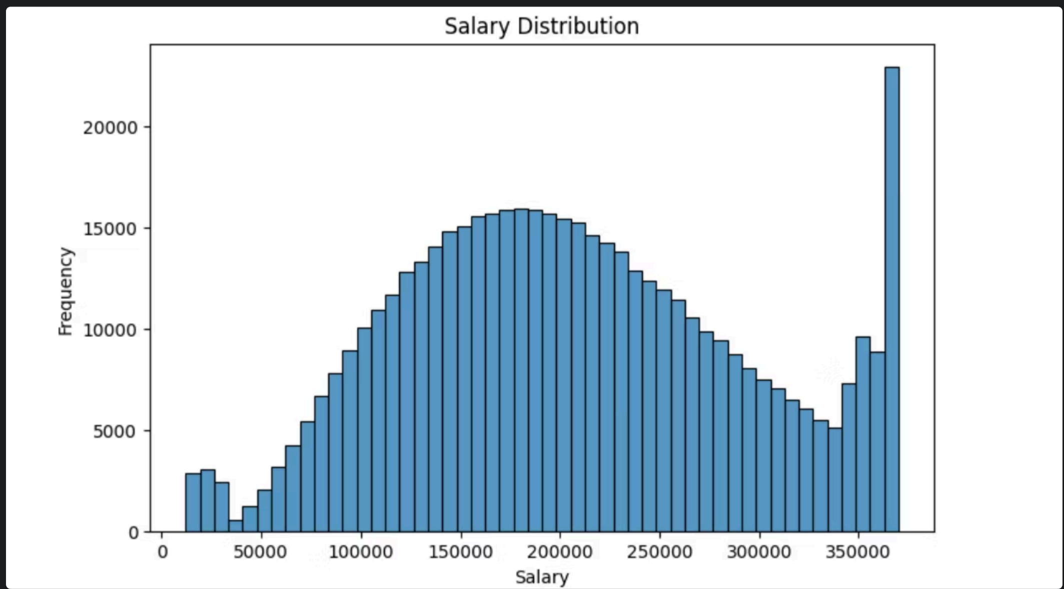


The job distribution is **heavily concentrated in United States**, while most other countries show **nearly equal and consistent counts**, indicating a **standardized global dataset with slight U.S. dominanc**

- Job opportunities are **evenly spread from 0 to 20 years**, but there is a **sharp spike at 20 years**, followed by a **drop beyond that**, suggesting **peak demand for experienced professionals with limited roles at very senior levels**.

Data Exploration (EDA — Part 2)





Key Insights from Visualizations

Top Occupations

Shows the most common job roles in the dataset, where roles like software engineer and data-related jobs appear more frequently, indicating high demand in the market.

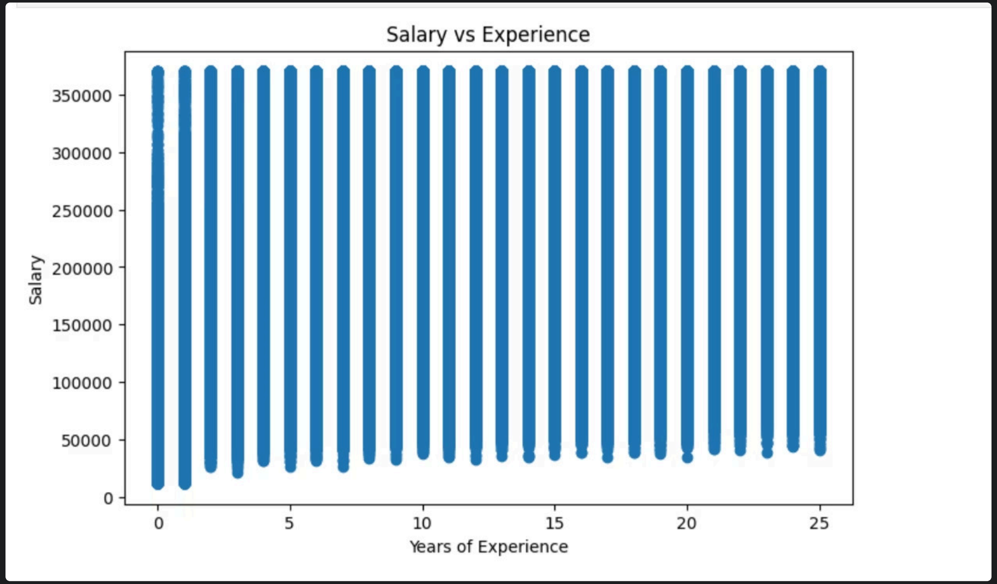
Salary Distribution

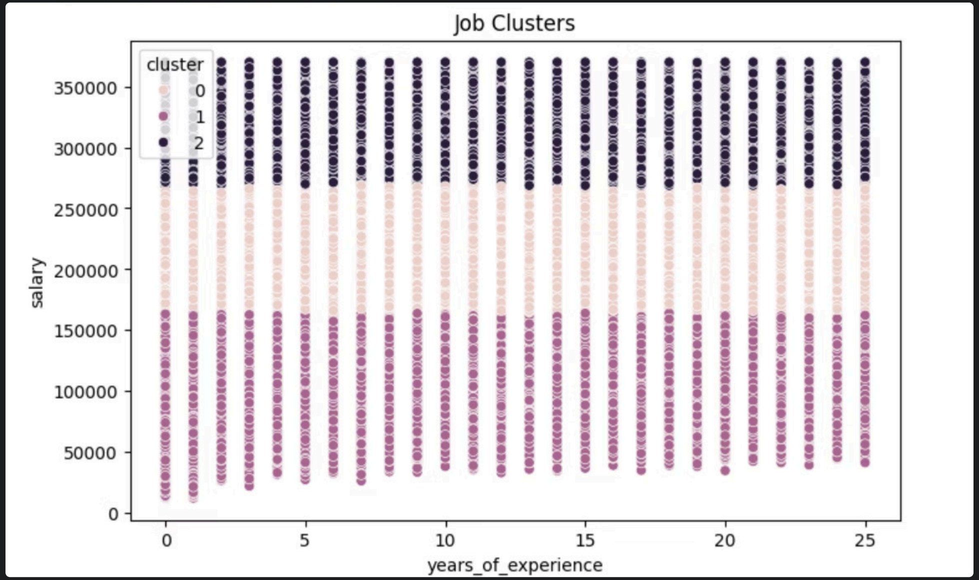
Shows how salaries are spread, forming a normal pattern where most people earn mid-range salaries and fewer people earn very low or very high salaries.

Balanced Records

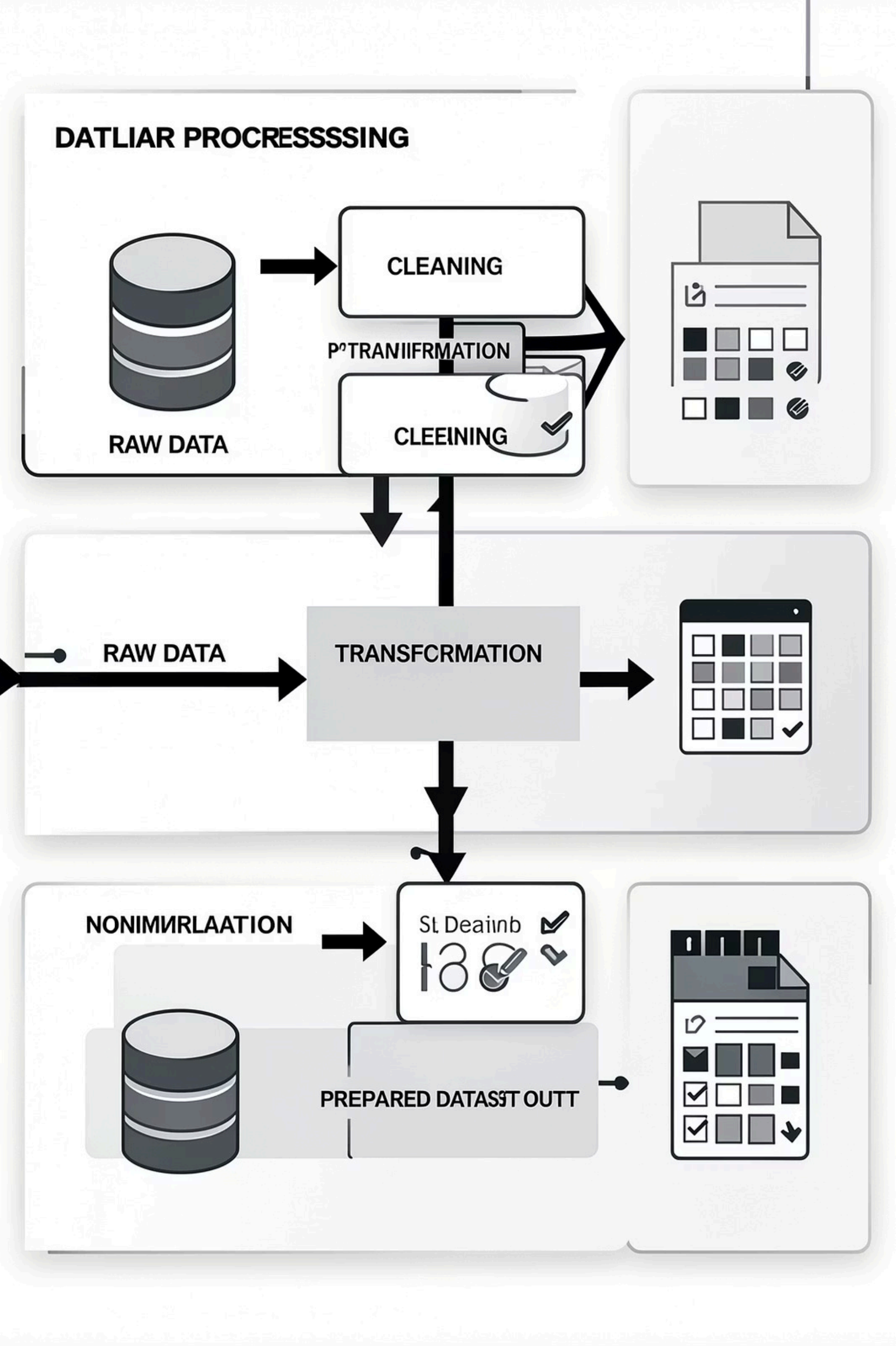
No extreme class imbalance in job categories, supporting fair pattern discovery.

Salary vs Experience and Job Clustering Analysis





The **Salary vs Experience graph** shows a positive relationship where salary increases as years of experience increase, meaning more experienced people generally earn higher salaries. The **Cluster graph** groups jobs into different clusters based on salary and experience, helping to identify similar types of jobs. These clusters show patterns in the job market, like high-salary experienced roles and low-salary beginner roles



Data Preprocessing

Raw data was transformed into a machine-learning-ready format through systematic encoding and scaling steps.

1

Clean Dataset

No missing values — dataset required no imputation or row removal.

2

Label Encoding

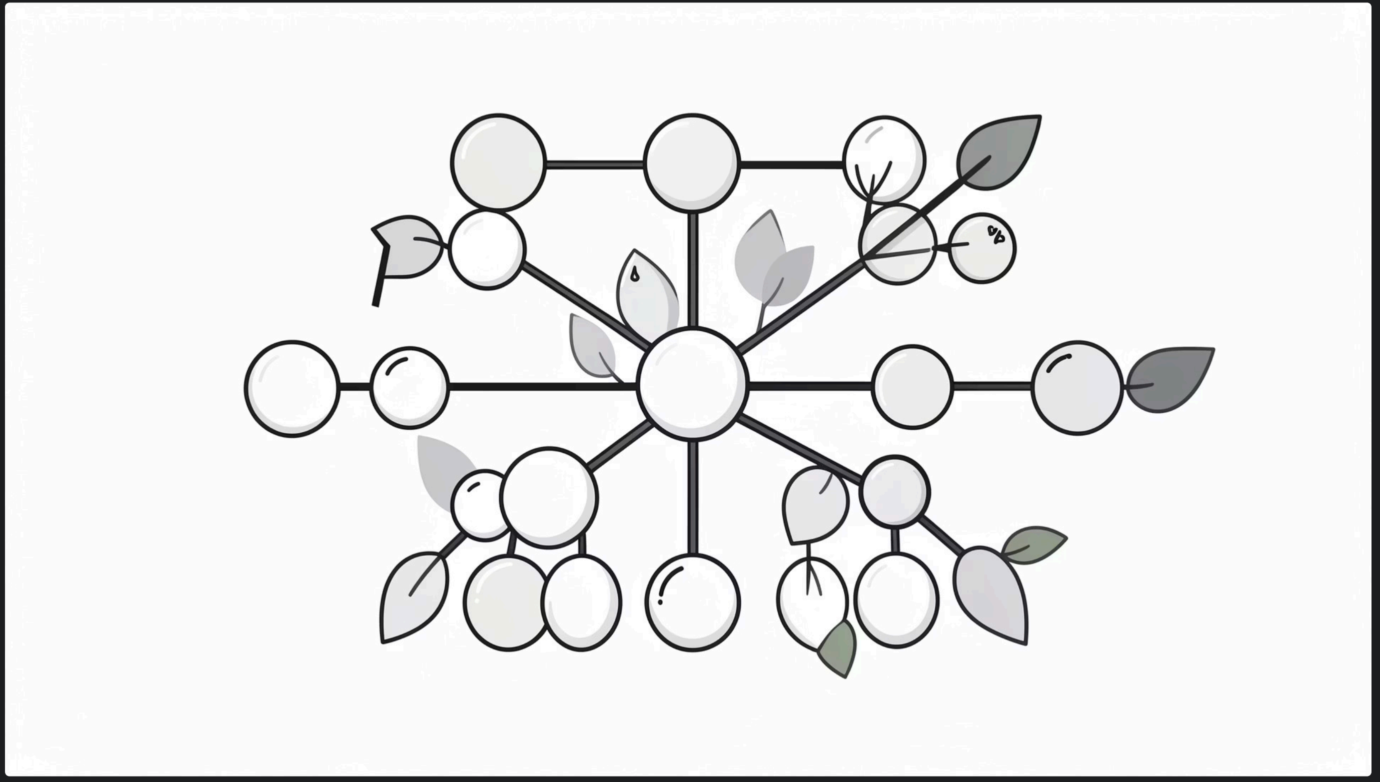
Categorical features (occupation, country, education) converted to numerical format.

3

Feature Scaling

StandardScaler applied to normalize numerical ranges and improve model convergence.

Data Mining Algorithms



Decision Tree Regressor

Goal: Predict salary based on job attributes like experience, education, and location.

- Interpretable, rule-based model
- Captures non-linear salary relationships
- Identifies most influential features

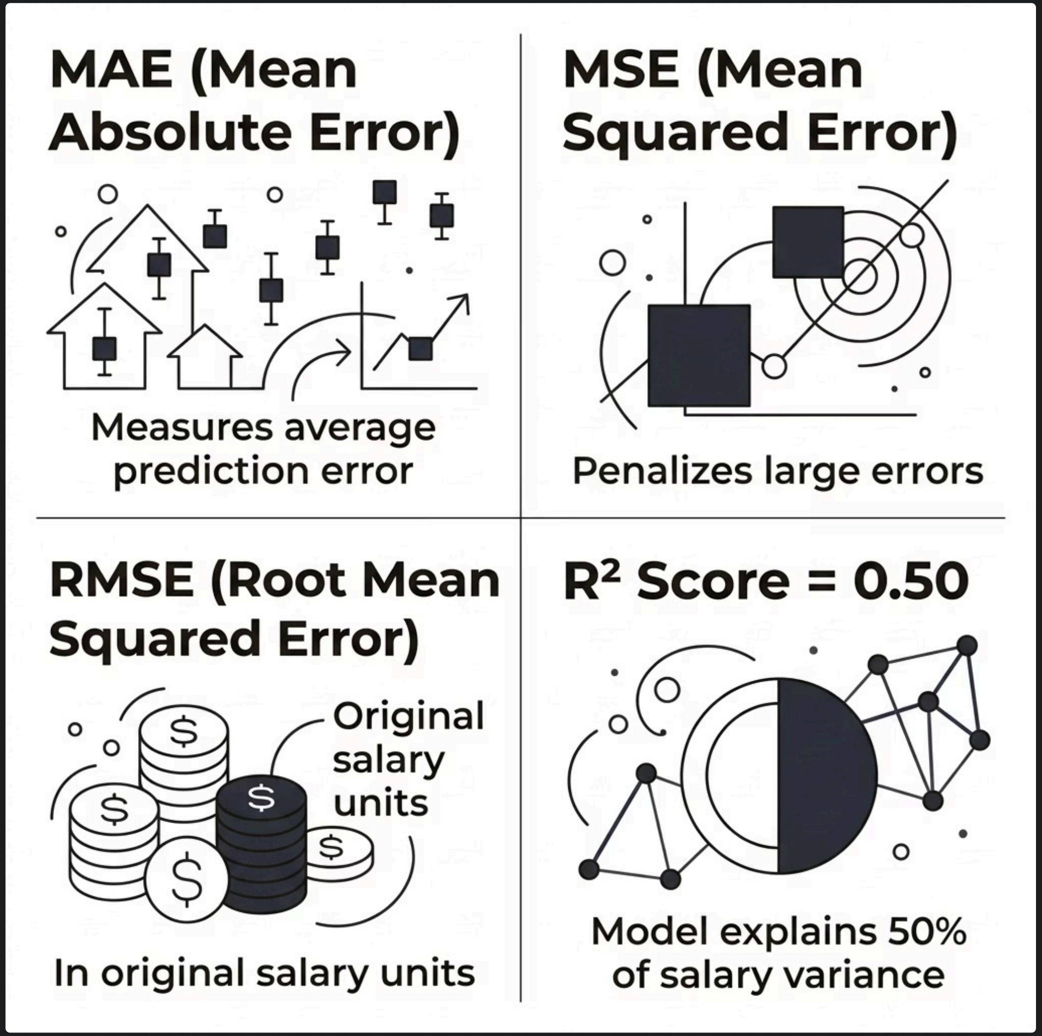
K-Means Clustering

Goal: Discover natural groupings of similar job profiles in the dataset.

- Unsupervised learning — no labels needed
- Groups jobs by salary, role, and experience
- Reveals hidden market segments

Model Evaluation

DECISION TREE REGRESSOR — RESULTS



What the Metrics Tell Us

An **R^2 score of 0.50** indicates that the model explains approximately half of the variance in salary — a moderate fit for a complex, real-world target variable.

> The model successfully identifies directional relationships between salary and features like experience, education, and job role — even where precise prediction is challenging.

Further tuning (depth limits, feature selection) could improve performance in future iterations.

Visualization & BI Insights

Salary vs. Experience

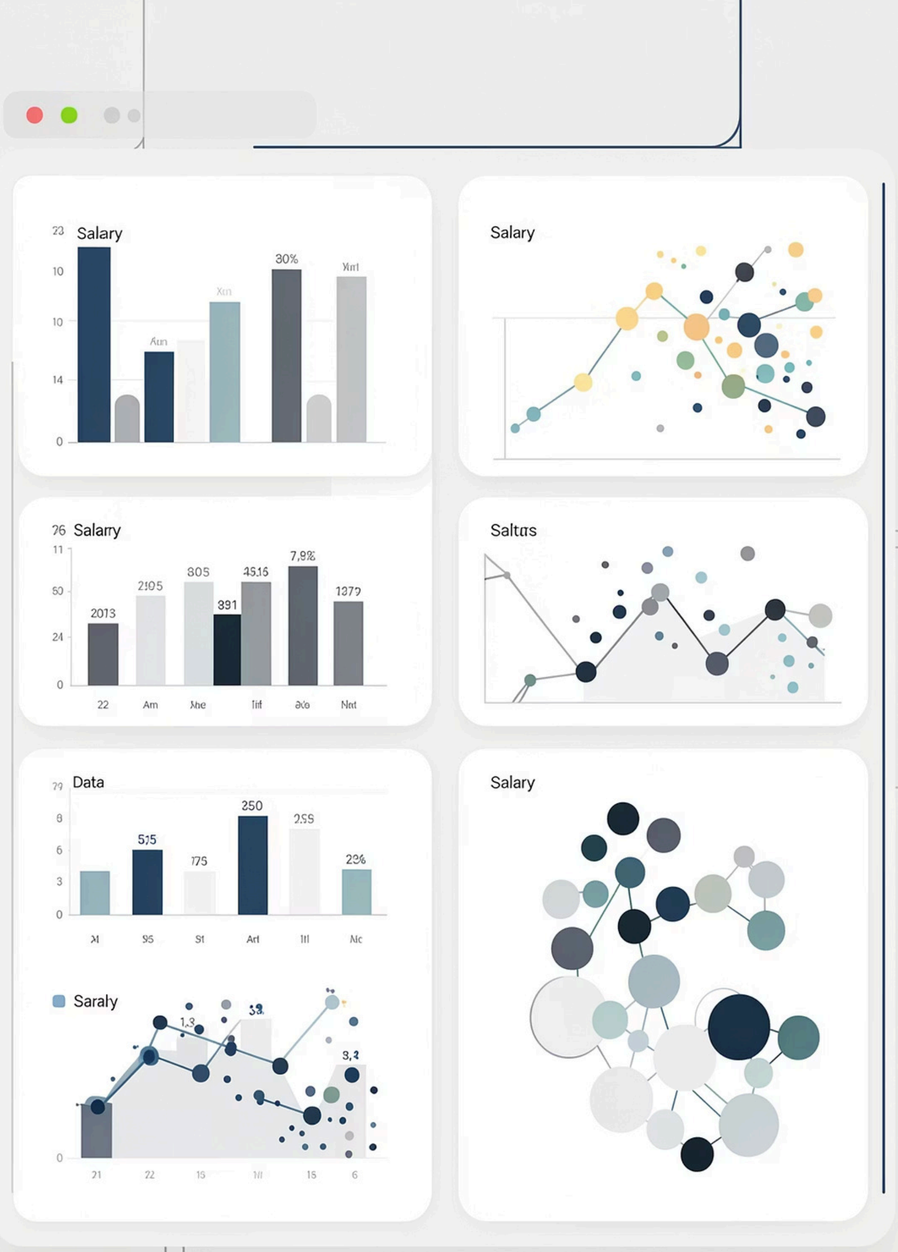
A clear positive correlation — compensation rises steadily as years of experience increase.

Salary by Education

Higher educational attainment consistently maps to higher salary bands across all job fields.

Job Clusters

K-Means groups jobs into distinct clusters — separating high-pay, mid-pay, and entry-level roles.



Made with GAMMA

Conclusion

Salary Drivers Identified

Experience, education level, and job role are the strongest predictors of salary in the global job market.

Power of Data Mining

Regression and clustering techniques together deliver both prediction and pattern discovery — complementary tools for workforce analytics.

Real-World Value

Findings benefit **job seekers** in benchmarking compensation and **companies** in designing competitive salary structures.

Global Perspective

Analyzing 500,000 records across geographies reveals universal workforce trends that transcend regional boundaries.

Data mining transforms raw job records into strategic intelligence — empowering smarter decisions for individuals and organizations alike.